

Linear Regression

Modelo: $p(y|\mathbf{x}, \mathbf{w}) = \mathcal{N}(y|\mathbf{w}^\top \mathbf{x}, \sigma^2)$

Predicción: $\hat{y} = \mathbb{E}[y|\mathbf{x}, \mathbf{w}] = \mathbf{w}^\top \mathbf{x}$

$\mathcal{L} = \text{MSE} = \frac{1}{N} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$

$\nabla_{\mathbf{w}} \mathcal{L} = \frac{2}{N} \sum_i (\mathbf{w}^\top \mathbf{x}_i - y_i) \mathbf{x}_i = \frac{2}{N} \mathbf{X}^\top (\mathbf{X}\mathbf{w} - \mathbf{y})$

Solución (MLE/LS): $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$

Multi-layer Perceptron (MLP)

$\mathbf{z}^{(\ell)} \in \mathbb{R}^{M^{(\ell)}}, \boldsymbol{\delta}^{(\ell)} = \mathbf{J}^{(\ell)} \boldsymbol{\delta}^{(\ell-1)} \in \mathbb{R}^{M^{(\ell)}}: \varepsilon$ de propagación en capa ℓ

$\mathbf{J}^{(\ell)} = \frac{\partial \mathbf{z}^{(\ell)}}{\partial \mathbf{z}^{(\ell-1)}} = \text{diag}(\varphi'(\ell)(\mathbf{a}^{(\ell)})) \cdot \mathbf{W}^{(\ell)}$

$\mathbf{W}^{(\ell)} \in \mathbb{R}^{M^{(\ell)} \times M^{(\ell-1)}}$

$\frac{\partial \mathcal{L}_i}{\partial \mathbf{W}^{(\ell)}} = \frac{\partial \mathcal{L}_i}{\partial \mathbf{z}^{(\ell)}} \frac{\partial \mathbf{z}^{(\ell)}}{\partial \mathbf{a}^{(\ell)}} \frac{\partial \mathbf{a}^{(\ell)}}{\partial \mathbf{W}^{(\ell)}}$

Back-propagation

Forward-pass

$\mathbf{z}^{(0)} \leftarrow \mathbf{x}_i$

for $\ell = 1 : L$ **do**

$\mathbf{a}^{(\ell)} \leftarrow \mathbf{W}^{(\ell)} \mathbf{z}^{(\ell-1)} + \mathbf{b}^{(\ell)}$

$\mathbf{z}^{(\ell)} \leftarrow \varphi^{(\ell)}(\mathbf{a}^{(\ell)})$

$\hat{\mathbf{y}}_i \leftarrow \mathbf{z}^{(L)}, \quad \mathcal{L}_i \leftarrow \mathcal{L}_i(\mathbf{y}_i, \hat{\mathbf{y}}_i)$

Backward-pass

$\boldsymbol{\delta}^{(L)} \leftarrow \varphi'^{(L)}(\mathbf{a}^{(L)}) \odot \frac{\partial \mathcal{L}_i}{\partial \mathbf{y}_i}$

$\frac{\partial \mathcal{L}_i}{\partial \mathbf{W}^{(L)}} \leftarrow \boldsymbol{\delta}^{(L)} \mathbf{z}^{(L-1)\top}$

$\frac{\partial \mathcal{L}_i}{\partial \mathbf{b}^{(L)}} \leftarrow \boldsymbol{\delta}^{(L)}$

for $l = L - 1 : 1$ **do**

$\boldsymbol{\delta}^{(l)} \leftarrow \varphi'^{(l)}(\mathbf{a}^{(l)}) \odot \mathbf{W}^{(l+1)\top} \boldsymbol{\delta}^{(l+1)}$

$\frac{\partial \mathcal{L}_i}{\partial \mathbf{W}^{(l)}} \leftarrow \boldsymbol{\delta}^{(l)} \mathbf{z}^{(l-1)\top}$

$\frac{\partial \mathcal{L}_i}{\partial \mathbf{b}^{(l)}} \leftarrow \boldsymbol{\delta}^{(l)}$

Activation Functions (φ)

Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}; \quad \sigma' = \sigma(1 - \sigma)$

Tanh: $\tanh' = 1 - \tanh^2$

ReLU: $\max(0, z); \quad \text{ReLU}' = \mathbb{I}[z > 0]$

Leaky ReLU: $\max(\alpha z, z), \quad \alpha \ll 1$

Softplus: $\log(1 + e^z)$

Identity: $\varphi(z) = z$

Softmax: $\sigma(\mathbf{z})_i = e^{z_i} / \sum_j e^{z_j}$

Ensemble Methods

Stacking: combina modelos heterogeneos

Bagging: paralelo, reduce varianza

- Random Forest:** + feature bagging.

Boosting: secuencial, reduce bias

- Grad. Boosting** (ajusta residuos con GD)
– XGBoost, LightGBM, CatBoost
- AdaBoost** (ponderación mayor a muestras mal clasificadas)

Distributions + MLE estimators

Gaussiana $\mathcal{N}(x|\mu, \sigma^2)$:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$\hat{\mu} = \frac{1}{N} \sum_i x_i \quad \hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \hat{\mu})^2$

Gaussiana Multivariada $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$:

$$\frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_i \mathbf{x}_i \quad \hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_i (\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^\top$

Bernoulli $\text{Ber}(x|\mu)$:

$$\mu^x (1 - \mu)^{1-x}$$

$\hat{\mu} = \frac{1}{N} \sum_i x_i$

Catagórica $\text{Cat}(y|\boldsymbol{\pi})$:

$$\prod_c \pi_c^{\mathbb{I}[y=c]}, \quad \sum_c \pi_c = 1$$

$\hat{\pi}_c = \frac{N_c}{N}$ (frecuencia de clase c)

Laplace $\text{Lap}(x|\mu, b)$:

$$\frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$

$\hat{\mu} = \text{median}(\{x_i\}) \quad \hat{b} = \frac{1}{N} \sum_i |x_i - \hat{\mu}|$

Binomial $\text{Binom}(k|n, p)$:

$$\binom{n}{k} p^k (1 - p)^{n-k}$$

$\hat{p} = \frac{1}{N} \sum_i \frac{k_i}{n}$

Logistic Regression (binary/multiclass)

Modelo: $p(y|\mathbf{x}, \boldsymbol{\theta}) = \text{Ber}(y|\sigma(\mathbf{w}^\top \mathbf{x} + b))$

Predicción (prob): $\mu_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$

$\mathcal{L} = \frac{1}{N} \sum_i \mathbb{H}_{ce}(y_i, \mu_i)$

$\nabla_{\mathbf{w}} \mathcal{L} = \frac{1}{N} \sum_i (\mu_i - y_i) \mathbf{x}_i = \frac{1}{N} \mathbf{X}^\top (\boldsymbol{\mu} - \mathbf{y})$

Modelo: $p(y|\mathbf{x}, \mathbf{w}) = \text{Cat}(y|\text{Softmax}(\mathbf{w}^\top \mathbf{x}))$

Predicción (prob): $\boldsymbol{\mu}_i = \text{Softmax}(\mathbf{w}^\top \mathbf{x}_i)$

$\mathcal{L} = \frac{1}{N} \sum_i \mathbb{H}_{ce}(\mathbf{y}_i, \boldsymbol{\mu}_i)$

$\nabla_{\mathbf{w}} \mathcal{L} = \frac{1}{N} \sum_i \mathbf{x}_i (\boldsymbol{\mu}_i - \mathbf{y}_i)^\top = \frac{1}{N} \mathbf{X}^\top (\mathbf{M} - \mathbf{Y})$
($\mathbf{M}$ matriz de predicciones, $\mathbf{Y}$ one-hot)

Naive Bayes

Naive Bayes Assumption: $x_d \perp x_{d'} \mid y = c$

Modelo:

$p(y = c|\mathbf{x}) \propto \pi_c \prod_{d=1}^D p(x_d|y = c, \boldsymbol{\theta}_{dc})$

$p(x_d|y = c, \boldsymbol{\theta}_{dc})$ según tipo de x_d :

- Binaria $\text{Ber}(x_d|\theta_{dc})$
- Catagórica = $\text{Cat}(x_d|\boldsymbol{\theta}_{dc})$
- Real = $\mathcal{N}(x_d|\mu_{dc}, \sigma_{dc}^2)$

Entrenamiento: MLE

Non-Parametric Methods

KNN-classifier:

$p(y = c|\mathbf{x}, \mathcal{D}) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(\mathbf{x}, \mathcal{D})} \mathbb{I}(y_i = c)$

Parzen Window Density Estimator:

$p(\mathbf{x}, \mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \mathcal{K}_h(\mathbf{x}, \mathbf{x}_i)$

NN Regularization + ∇ vanishing sol.

Weight Decay (ℓ_2 regularization)

Dropout: $\theta_{ji}^{(\ell)} = w_{ji}^{(\ell)} \epsilon_i^{(\ell)}, \epsilon \sim \text{Ber}(1 - p)$

Al predecir: $w_{ij}^{(\ell)} = \theta_{ij}^{(\ell)} \mathbb{E}[\epsilon_i^{(\ell)}] = \theta_{ij}^{(\ell)} (1 - p)$

Early Stopping: frenar cuando $\uparrow \mathcal{L}_{\text{val}}$

ℓ_1 regularization

Batch/Layer Normalization: normalizar por mini-batch (μ_B, σ_B^2) o por muestra sobre features ($\mu^{(\ell)}, \sigma^{2(\ell)}$)

$\hat{z}^{(\ell)} = \frac{z^{(\ell)} - \mu}{\sqrt{\sigma^2 + \epsilon}}, \quad \tilde{z}^{(\ell)} = \gamma \hat{z}^{(\ell)} + \beta$

Glorot/Xavier initialization

$w_{ij} \sim \mathcal{N}(0, \sigma^2)$ con $\sigma^2 = 2/(n_{\text{in}} + n_{\text{out}})$

φ sin saturación: ReLU, Leaky ReLU, etc

Gradient clipping: $\mathbf{g} = \min\left(0, \frac{c}{\|\mathbf{g}\|}\right) \mathbf{g}$

Maximum A Posteriori (MAP)

$\mathbf{w}^* = \arg \max_{\mathbf{w}} p(\mathbf{w}|\mathcal{D}) = \arg \max_{\mathbf{w}} p(\mathcal{D}|\mathbf{w})p(\mathbf{w})$

- ℓ_2 : $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \lambda^{-1} \mathbf{I}) \rightarrow \mathcal{L} + \lambda \|\mathbf{w}\|^2$
- ℓ_1 : $p(\mathbf{w}) = \text{Lap}(\mathbf{w}|\mathbf{0}, \lambda^{-1}) \rightarrow \mathcal{L} + \lambda \|\mathbf{w}\|_1$

Bayes

$$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w}) p(\mathbf{w})}{p(\mathcal{D})} \propto p(\mathcal{D}|\mathbf{w}) p(\mathbf{w})$$

posterior $\propto$ *likelihood* $\times$ *prior*

Information Theory

Entropy: $\mathbb{H}(\mathbf{p}) = -\sum_i p_i \log p_i$

Cross-E: $\mathbb{H}_{ce}(\mathbf{p}, \mathbf{q}) = -\sum_i p_i \log q_i$

Binary Cross-E:

$\mathbb{H}_{ce}(p, q) = -[p \log q + (1 - p) \log(1 - q)]$

Joint-E: $\mathbb{H}(X, Y) = -\sum p(x, y) \log p(x, y)$

Conditional-E: $\mathbb{H}(Y|X) = \mathbb{H}(X, Y) - \mathbb{H}(X)$

Relative-E (KL divergence)

$\mathbb{H}(p||q) = \sum_i p_i \log \frac{p_i}{q_i} = \mathbb{H}(p||q) - \mathbb{H}(p)$

Mutual Information

$I(X; Y) = \mathbb{H}(p(x, y) || p(x)p(y))$

$I(X; Y) = 0 \Leftrightarrow X \perp Y$

Convexity

$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y})$

1er orden: $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$

2do orden: $\nabla^2 f(\mathbf{x}) \succeq 0 \forall \mathbf{x}$ (Hessiano SDP)

GDA

Modelo: $p(y = c|\mathbf{x}, \mathbf{w}) \propto \pi_c p(\mathbf{x}|y = c, \mathbf{w})$

Class-cond: $p(\mathbf{x}|y = c, \mathbf{w}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$

Prior: $\pi_c = p(y = c|\mathbf{w})$

Entrenamiento: MLE ($\hat{\pi}, \hat{\boldsymbol{\mu}}_c, \hat{\boldsymbol{\Sigma}}_c$)

MAP: $\hat{\boldsymbol{\Sigma}}_{\text{map}} = \lambda \text{diag}(\hat{\boldsymbol{\Sigma}}) + (1 - \lambda) \hat{\boldsymbol{\Sigma}}$

QDA ($\boldsymbol{\Sigma}_c$ libre por clase)

Frontera: $\mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c_0 = 0$

- $\mathbf{A} = \frac{1}{2} (\boldsymbol{\Sigma}_{c'}^{-1} - \boldsymbol{\Sigma}_c^{-1}), \quad \mathbf{b} = \boldsymbol{\Sigma}_c^{-1} \boldsymbol{\mu}_c - \boldsymbol{\Sigma}_{c'}^{-1} \boldsymbol{\mu}_{c'}$
- $c_0 = \log \frac{\pi_c}{\pi_{c'}} - \frac{1}{2} \log \frac{|\boldsymbol{\Sigma}_c|}{|\boldsymbol{\Sigma}_{c'}|} - \frac{1}{2} (\boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}_c^{-1} \boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'}^\top \boldsymbol{\Sigma}_{c'}^{-1} \boldsymbol{\mu}_{c'})$

LDA ($\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma} \forall c$)

Frontera: $\mathbf{w}^\top \mathbf{x} + w_0 = 0$

- $\mathbf{w} = \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'})$
- $w_0 = \log \frac{\pi_c}{\pi_{c'}} - \frac{1}{2} (\boldsymbol{\mu}_c + \boldsymbol{\mu}_{c'})^\top \mathbf{w}$

CART

$\mathcal{L}(\boldsymbol{\theta}) = \sum_i \ell(y_i, f(\mathbf{x}_i; \boldsymbol{\theta}))$

Split (nodo i):

$\arg \min_j \min_t \frac{|\mathcal{D}_i^L|}{|\mathcal{D}_i|} C^L(\mathcal{D}_i^L) + \frac{|\mathcal{D}_i^R|}{|\mathcal{D}_i|} C^R(\mathcal{D}_i^R)$

Regresión: $C(\mathcal{D}_i) = \text{MSE}$

Clasificación: $C(\mathcal{D}_i) = G_i = \sum_c \hat{\pi}_{ic} (1 - \hat{\pi}_{ic})$
ó $C(\mathcal{D}_i) = H_i = \mathbb{H}(\hat{\boldsymbol{\pi}}_i) = -\sum_{ic} \hat{\pi}_{ic} \log(\hat{\pi}_{ic})$

Perceptron

$\hat{y} = \mathbb{I}(\mathbf{w}^\top \mathbf{x} + b > 0) \rightarrow$ (determinístico)

Update: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta (\hat{y}_i - y_i) \mathbf{x}_i$

Optimization

GD: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t \nabla_{\mathbf{w}} \mathcal{L}|\mathbf{w}_t$

SGD: GD con una muestra ($\mathbf{x}_i, y_i$) por update

Mini-batch: $\mathcal{B} \subset \mathcal{D}$ por update

Momentum: acumula gradientes pasados

$\mathbf{m}_t \leftarrow \beta \mathbf{m}_{t-1} + \nabla \mathcal{L}|\mathbf{w}_t \quad \mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t \mathbf{m}_t$

Newton: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t \mathbf{H}_t^{-1} \nabla_{\mathbf{w}} \mathcal{L}$

LR Schedule: $\downarrow \eta_t$ por iteración

- Lineal: $\eta_t = (1 - \frac{t}{T}) \eta_0 + (\frac{t}{T}) \eta_T$
- Potencia: $\eta_t = \eta_0 (1 + \frac{t}{s})^c, \quad c, s$ params
- Exponencial: $\eta_t = \eta_0 \cdot c^{(t/s)}, \quad c, s$ params

AdaGrad: η_t adaptativo por parámetro

Adam: Momentum + AdaGrad (η y m por parámetro)

Regression Metrics

$e_i \triangleq y_i - \hat{y}_i; \text{RSS} \triangleq \sum_i e_i^2; \text{TSS} \triangleq \sum_i (y_i - \bar{y})^2$

$\text{MSE} = \frac{1}{N} \text{RSS}$

$\text{RMSE} = \sqrt{\text{MSE}}$

$\text{MAE} = \frac{1}{N} \sum_i |e_i|$

$\mathbf{R}^2 \triangleq 1 - \frac{\text{RSS}}{\text{TSS}} \in [0, 1]$ (varianza explicada)

Classification Metrics

$\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN} \quad \text{Recall} = \frac{TP}{TP+FN}$

$\text{Precision} = \frac{TP}{TP+FP} \quad \text{FPR} = \frac{FP}{FP+TN}$

$\text{FDR} = \frac{FP}{FP+TP}$

$\mathbf{F}_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP+FP+FN}$

ROC Curve: FPR vs TPR

PR Curve: Recall vs Precision

Linear Algebra

SDP: $\mathbf{A} \succeq 0 \Leftrightarrow \mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0 \forall \mathbf{x}$.

$\mathbf{A}^\top \mathbf{A}, \mathbf{A} \mathbf{A}^\top \succeq 0$ siempre

Cond: $\kappa(\mathbf{A}) = \sigma_{\text{max}}/\sigma_{\text{min}}, \kappa(\mathbf{A}^\top \mathbf{A}) = \kappa(\mathbf{A})^2$

Precisión: $\boldsymbol{\Lambda} \triangleq \boldsymbol{\Sigma}^{-1}$

Derivadas conocidas

- $\nabla_{\mathbf{x}}(\mathbf{x}^\top \mathbf{A} \mathbf{x}) = 2\mathbf{A} \mathbf{x}$ ($\mathbf{A}$ simétrica)
- $\nabla_{\mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}$
- $\nabla_{\boldsymbol{\Sigma}} \log |\boldsymbol{\Sigma}| = \boldsymbol{\Sigma}^{-\top}$

Traza: $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top), \text{tr}(\mathbf{A} \mathbf{B}) = \text{tr}(\mathbf{B} \mathbf{A})$

Jacobiano: $\mathbf{J} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \in \mathbb{R}^{m \times n}, J_{ij} = \frac{\partial f_i}{\partial x_j}$

Si $m = 1 \implies \mathbf{J} = \nabla f^\top$.